

Md Saqlain Khan

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Professional Summary

Software Engineer with a background in Architecture, passionate about building software that connects engineering, computational design, artificial intelligence, and the built environment. Experienced in designing and shipping full-stack applications, engineering automation platforms, and AI-powered tools used in production engineering workflows.

Currently leading technical delivery for production Intel engineering software at AtkinsRéalis, coordinating development across a small engineering team while contributing hands-on to architecture, implementation, production support, and AI integration. I enjoy building products from concept to deployment, combining strong engineering fundamentals with product thinking, scalability, and user-focused design.

My interests lie at the intersection of software engineering, computational geometry, AI, and real-world engineering systems, where software has a direct impact on how physical infrastructure is designed, built, and operated.

Selected Highlights

- Leading technical delivery for a production Intel engineering platform while coordinating development across a team of four engineers.
- Built and launched ObjectTracer, a real-time multi-domain tracking platform serving 200+ active users.
- Optimized visualization architecture capable of rendering more than 40,000 live objects in real time.
- Developed AI-powered engineering applications integrating Large Language Models into enterprise workflows.
- Background spanning Architecture, Computational Design, Full Stack Engineering, and Artificial Intelligence.

Education

National Institute of Technology Patna (Institute of National Importance)

GPA: 3.66 / 4.0

Bachelor of Architecture

2018 – 2023

Experience

Software Development Engineer 1 (Intel Projects)

Dec 2025 – Present

AtkinsRéalis, Bangalore

- Lead technical delivery for a production engineering platform supporting Intel projects, coordinating development across a team of four engineers while contributing hands-on to architecture, implementation, and production support.
- Design and develop scalable full-stack applications using modern web technologies, enabling engineering automation and AI-powered workflows.
- Drive technical decisions around software architecture, maintainability, performance, and long-term scalability.
- Mentor engineers by resolving technical challenges, guiding implementation, and helping unblock development throughout the project lifecycle.
- Lead investigation and resolution of critical production issues, performing root cause analysis and coordinating reliable fixes.
- Collaborate with engineering, product, and business stakeholders to prioritize features, manage technical trade-offs, and deliver customer-focused solutions.
- Integrate Large Language Models and AI capabilities into enterprise engineering workflows to improve productivity and automate repetitive engineering tasks.

Assistant Software Engineer

Mar 2025 – Nov 2025

AtkinsRéalis, Bangalore

- Developed enterprise engineering applications with a focus on scalability, maintainability, and reliability.
- Built desktop applications using C#, WPF, MVVM, and SQL Server to support engineering workflows.
- Improved application performance through asynchronous programming, optimized database operations, and code refactoring.
- Worked closely with multidisciplinary engineering teams to deliver production software in Agile environments.

Graduate Software Engineer

Mar 2024 – Feb 2025

AtkinsRéalis, Bangalore

- Developed full-stack engineering applications using React, Node.js, Autodesk Platform Services (APS), and BIM360.
- Built backend services and event-driven workflows supporting engineering automation and project data synchronization.
- Developed engineering dashboards and visualization tools integrating BIM and infrastructure data.
- Built secure WPF desktop applications with SQL Server backends for enterprise engineering teams.
- Collaborated across frontend, backend, and cloud teams to rapidly prototype and deliver engineering solutions.

Software Engineer Trainee

Jul 2023 – Feb 2024

AtkinsRéalis, Bangalore

- Automated architectural and engineering workflows using C# and the Revit API, reducing manual effort by approximately 70%.
- Developed productivity tools for engineering model validation, reporting, and workflow automation.
- Built reusable software components that improved maintainability across multiple engineering applications.

Architectural Intern

Jun 2022 – Dec 2022

Gensler, Bangalore

- Contributed to large-scale workplace and hospitality projects, gaining practical exposure to multidisciplinary design and construction workflows.
- Developed computational design and engineering automation tools using Rhino, Grasshopper, Revit, Dynamo, and Rhino APIs.
- Built procedural geometry generation tools that automated repetitive design tasks and improved workflow efficiency.
- Collaborated with architects and engineers to support digital design workflows across multiple project stages.
- This experience inspired my transition into software engineering, combining architecture, computation, and technology to solve engineering challenges.

Projects

ObjectTracer : Real-Time Multi-Domain Tracking Platform

- Designed, developed, and launched a real-time tracking platform for aircraft, satellites, ships, and other live objects across aviation, maritime, and space domains.
- Serves more than 200 active users and continues to grow through iterative development, user feedback, analytics-driven improvements, and technical SEO.
- Optimized the rendering architecture to visualize more than 40,000 live objects while maintaining smooth interactive performance.
- Built scalable backend services in Go and an interactive React frontend with modular APIs capable of supporting multiple real-time data sources.
- Owned the complete product lifecycle, including architecture, development, deployment, analytics, search optimization, monitoring, and continuous feature delivery.
- **Tech Stack:** React, Golang, REST APIs, Railway, Vercel
- **Live:** objecttracer.com
- **GitHub:** github.com/starman011/flightspace

AI Engineering Platform for Infrastructure Workflows (Intel Internal)

- Developed an AI-powered engineering platform integrating Large Language Models into enterprise infrastructure workflows.
- Built backend services supporting intelligent engineering assistance, workflow automation, document processing, and AI-driven task execution.
- Designed intuitive React interfaces enabling engineers to interact naturally with AI capabilities across multiple engineering workflows.
- Developed reusable AI components that could be integrated across different engineering applications and business use cases.
- Collaborated closely with engineering stakeholders to deliver AI capabilities that improved engineering productivity and reduced repetitive manual work.
- **Tech Stack:** React, Golang, LLM Integration, AI Workflow Automation

Interactive Model Customization and Automated Rendering Platform

- Developed a full-stack MERN application enabling interactive model customization and automated rendering workflows.
- Integrated Unity APIs to automate rendering and video generation, reducing manual effort by approximately 90%.
- Designed scalable backend services for asset management, rendering automation, and workflow orchestration.
- Delivered an intuitive web experience that significantly streamlined the rendering workflow for end users.

- **Tech Stack:** MongoDB, Express.js, React, Node.js, Unity APIs

Technical Skills

Programming Languages: Python, Go, C#, C++, TypeScript, JavaScript, SQL, Java

Full Stack Development: React.js, Node.js, Express.js, ASP.NET Core, REST APIs, WebSockets

Artificial Intelligence: Large Language Models (LLMs), AI Workflow Automation, Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Integration

Engineering Software: Autodesk Platform Services (APS), Revit API, Rhino, Grasshopper, Dynamo, BIM Workflows

Cloud & DevOps: Azure, Docker, CI/CD, Railway, Vercel, GitHub Actions

Databases: Microsoft SQL Server, MongoDB, SQLite

Software Engineering: System Design, Distributed Systems, Event-Driven Architecture, Performance Optimization, Multithreading

Tools: Git, GitHub, Visual Studio, VS Code, Power BI

Additional Information

Interests: Engineering Software, Computational Design, Artificial Intelligence, Real-Time Visualization, Product Development, Robotics, Physical AI, Architecture